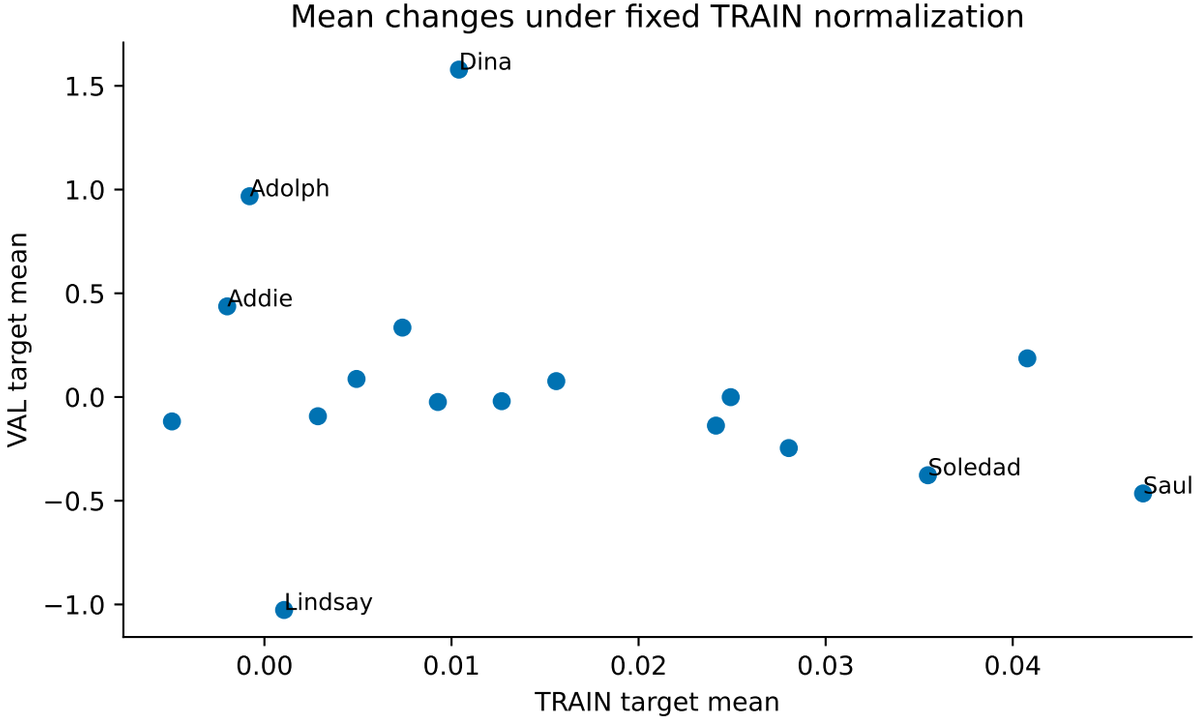
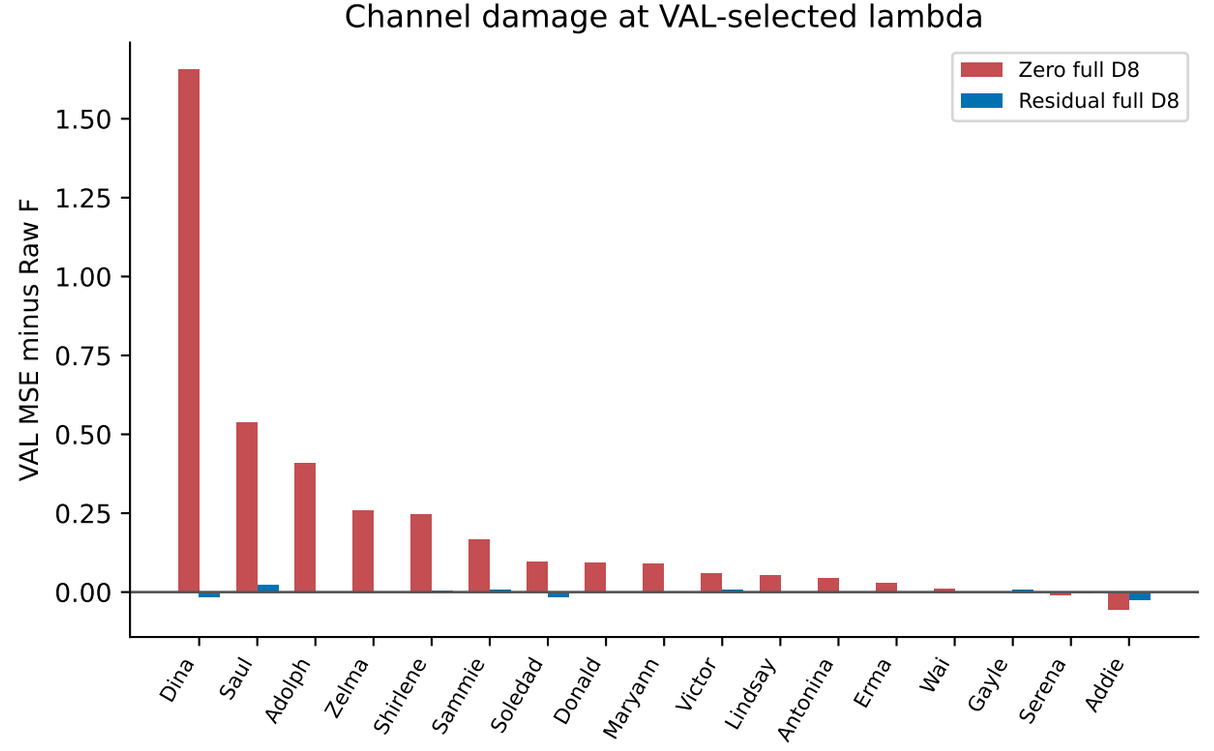
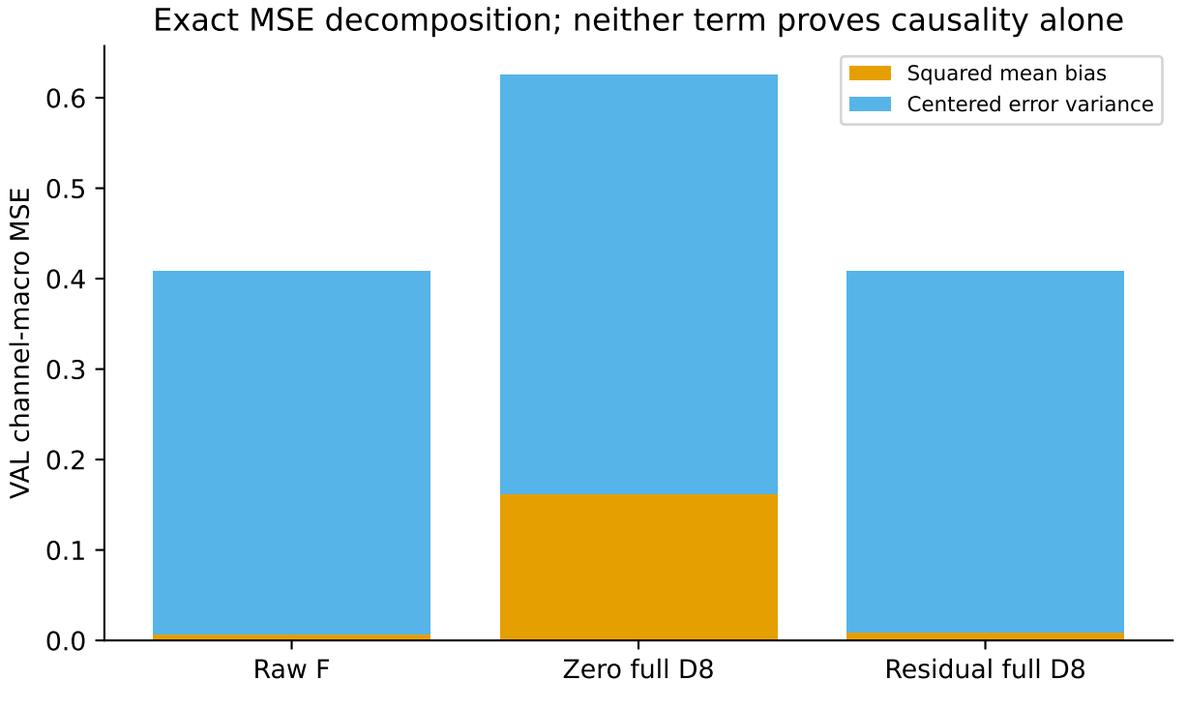
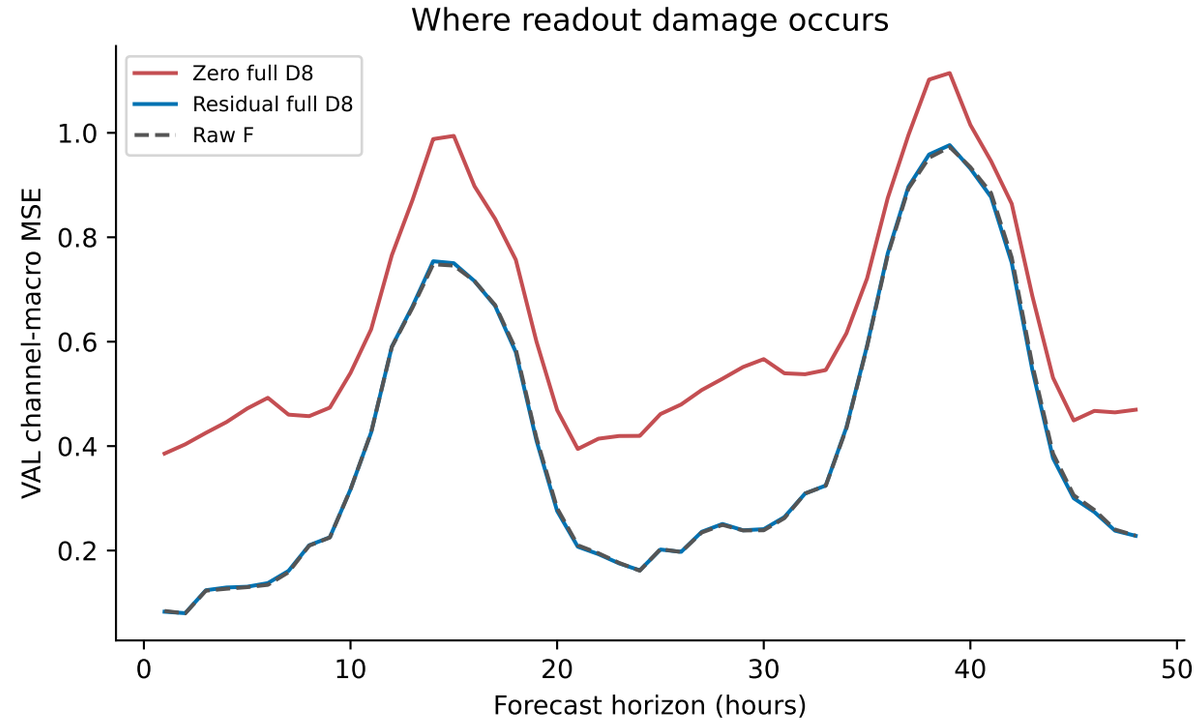

Mean changes under fixed TRAIN normalization
Dina
Adolph
Addie
Soledad
Saul
Lindsay
VAL target mean
TRAIN target mean
1.5
1.0
0.5
0.0
−0.5
−1.0
0.00
0.01
0.02
0.03
0.04


Channel damage at VAL-selected lambda
Zero full D8
Residual full D8
VAL MSE minus Raw F
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Dina
Saul
Adolph
Zelma
Shirlene
Sammie
Soledad
Donald
Maryann
Victor
Lindsay
Antonina
Erma
Wai
Gayle
Serena
Addie


Exact MSE decomposition; neither term proves causality alone
Squared mean bias
Centered error variance
VAL channel-macro MSE
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Raw F
Zero full D8
Residual full D8


Where readout damage occurs
Zero full D8
Residual full D8
Raw F
VAL channel-macro MSE
1.0
0.8
0.6
0.4
0.2
0
10
20
30
40
50
Forecast horizon (hours)